Fixed Point Theorems of Brouwer and Schauder

Chapter 5, Section 5.3: Pages 309–328

Lecture Notes

July 14, 2026

Key Topics:

- Nonexpansive and Asymptotically Nonexpansive Mappings
- Brouwer's Fixed Point Theorem
- Schauder's Fixed Point Theorem
- Applications and Extensions

Outline

Basic Definitions: Contractive Mappings

Definition 5.23 (Pseudocontractive Mapping)
Let K be a nonempty subset of a normed linear space X. A mapping $F : K \to X$ is said to be **pseudocontractive** if for each $k > 0$ and for all $x, y \in K$:

$$\|x - y\| \leq \|(1 + k)(x - y) - k(Fx - Fy)\| .$$

In a real Hilbert space, this is equivalent to:

$$\|Fx - Fy\|^2 \leq \|x - y\|^2 + \|(I - F)(x) - (I - F)(y)\|^2 \quad \forall x, y \in K.$$

Alternative form:

$$(Fx - Fy, x - y) \leq \|x - y\|^2 \quad \forall x, y \in X.$$

Significance: Pseudocontractive mappings are important in nonlinear analysis due to their connection with accretive operators.

Nonexpansive Mappings

Definition: A mapping $T : K \to K$ is **nonexpansive** if

$$\|Tx - Ty\| \leq \|x - y\| \quad \forall x, y \in K.$$

Key Properties:

1. Lipschitz constant $= 1$
2. Does not require contraction
3. Fixed point may not be unique
4. Exists under restrictive conditions (e.g., star-shaped sets in Hilbert spaces)

Theorem 5.40 (Kirk) Let $\mathcal{H}$ be a real Hilbert space and K a closed star-shaped subset of $\mathcal{H}$. Suppose $F : K \to \mathcal{H}$ is nonexpansive with $F(\partial K) \subset K$ and $\{F^n(x_0)\}$ is bounded. Then F has a fixed point.

Asymptotically Nonexpansive Mappings

Definition 5.24 A mapping $T : K \to K$ is **asymptotically nonexpansive** if

$$\|T^n x - T^n y\| \leq k_n \|x - y\| \quad \forall x, y \in K \quad \text{and} \quad \lim_{n \to \infty} k_n = 1.$$

Definition 5.25 A mapping $T : K \to K$ is **asymptotically regular** if for any $x \in K$,

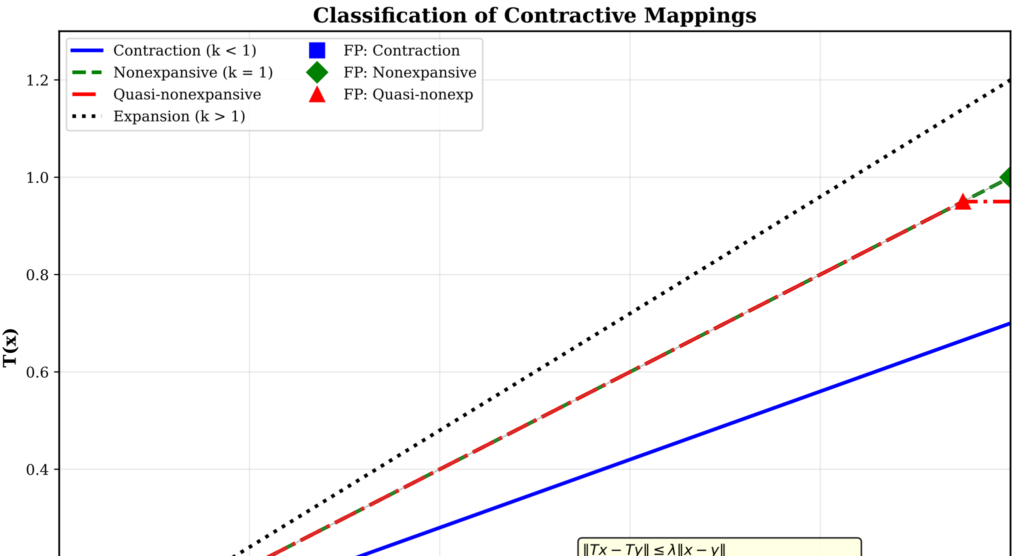
$$\left\| T^n x - T^{n+1} x \right\| \to 0 \quad \text{as } n \to \infty.$$

Theorem 5.42 (Goebel and Kirk, 1972) Let K be nonempty, closed, convex and bounded subset of a uniformly convex Banach space and let T be an asymptotically nonexpansive mapping of K into itself. Then T has a fixed point in K.

This is a major extension of the Brouwer fixed point theorem to infinite-dimensional settings.

Comparison of Mapping Types

Visual Illustration



Brouwer's Fixed Point Theorem

Fundamental Result

Theorem 5.48 (Brouwer's Fixed Point Theorem)
The closed unit ball $\mathbb{B}^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ of $\mathbb{R}^n$ has the **fixed point property**.
Alternatively: Let $f : \mathbb{B}^n \to \mathbb{B}^n$ be a continuous function. Then f has a fixed point $\bar{x} \in \mathbb{B}^n$.

Historical Note:

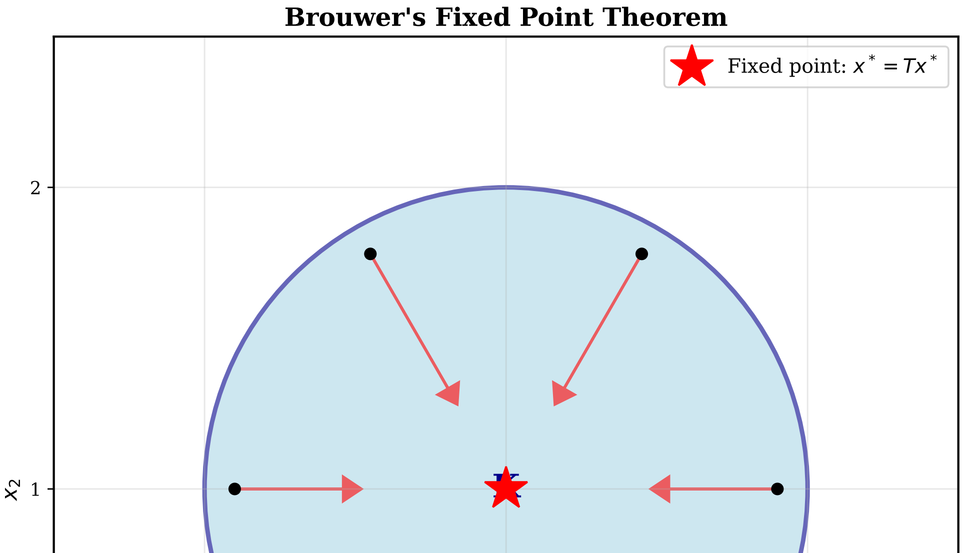
- Brouwer: 1910 (first complete proof for $n = 3$)
- Poincaré: 1886 (different version)
- Brouwer: 1912 (proof for arbitrary n)

Limitations:

- Does NOT provide computational scheme
- Does NOT guarantee uniqueness
- Limited to finite dimensions (needs modification for infinite dimensions)

Brouwer's Theorem: Geometric Interpretation

Topological Proof Sketch



Lemma 5.7: Topological Foundation I

Lemma 5.7 The set $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ is not a retract of $\mathbb{B}^n$.

Proof Sketch (using Algebraic Topology):
A retraction $r : \mathbb{B}^n \to S^{n-1}$ would induce a homomorphism:

$$r_* : H_{n-1}(\mathbb{B}^n) \to H_{n-1}(S^{n-1})$$

- Natural injection: $j : S^{n-1} \to \mathbb{B}^n$ induces j_*
- Composition: $(r \circ j)_* = r_* \circ j_*$ is the identity on $H_{n-1}(S^{n-1})$
- But: $H_{n-1}(\mathbb{B}^n) = 0$ (ball is contractible)
- While: $H_{n-1}(S^{n-1}) = \mathbb{Z}$ (sphere has nontrivial homology)
- Contradiction!

Lemma 5.7: Topological Foundation II

Analytical Proof (using Differential Forms):
Associate to C^2 function $h : \mathbb{B}^n \to \mathbb{B}^n$ the exterior form:

$$\omega_h = h_1 dh_2 \wedge \cdots \wedge dh_n$$

By Stokes theorem:

$$\int_{S^{n-1}} \omega_h = \int_{\mathbb{B}^n} d\omega_h = \int_{\mathbb{B}^n} \det[J_h(x)]dx$$

If retraction r of class C^2 existed, then $\det[J_r] \equiv 0$, so:

$$\int_{S^{n-1}} \omega_r = 0$$

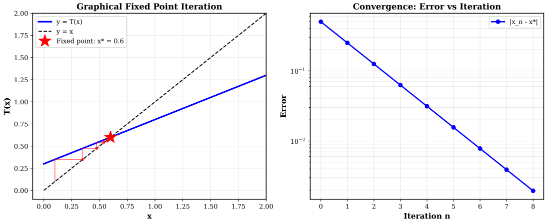
But $r|_{S^{n-1}} = i_{S^{n-1}}$ (identity), so integral must equal volume of S^{n-1}. Contradiction!

Application: Numerical Example

Fixed Point Iteration

Problem: Find fixed point of $f(x) = 0.5x + 0.3$ on $\mathbb{B}^1 = [-1, 1]$.

```python
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return 0.5 * x + 0.3

# Fixed point iteration
x_init = 0.1
x_seq = [x_init]
for i in range(10):
    x_new = f(x_seq[-1])
    x_seq.append(x_new)

x_seq = np.array(x_seq)

# Exact fixed point: x* = 0.5*x* + 0.3
x_star = 0.6

print(f"Iterations: {x_seq}")
print(f"Fixed point: {x_star}")
print(f"Final error: {abs(x_seq[-1] - x_star)}")
```



Output:

- Iterations converge to $x^* = 0.6$
- Exponential error decay
- By Brouwer: $\exists x^* : f(x^*) = x^*$

Schauder's Fixed Point Theorem

Extension to Infinite Dimensions

Theorem 5.54 (Schauder's Fixed Point Theorem)
Let X be a Banach space, K a nonempty, closed and convex subset of X, and $T : K \to K$ a continuous mapping. If $T(K)$ is relatively compact in K, then T has a fixed point in K.

Corollary 5.11: If K is compact and convex, and $T : K \to K$ is continuous, then T has a fixed point.

Key Advantage over Brouwer:

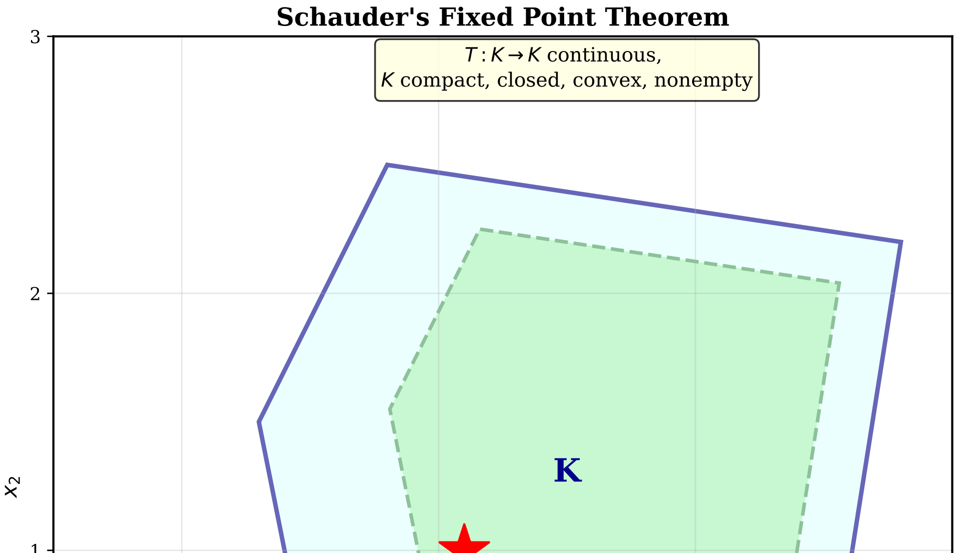
- Works in **infinite-dimensional** spaces
- Applies to more general settings (Fréchet spaces, etc.)
- No restriction to finite balls

Comparison with Brouwer:

Property	Brouwer	Schauder
Finite dimension	Yes	Yes
Infinite dimension	No	**Yes**
Compact domain	No (ball OK)	**Yes**
Convex domain	Yes	Yes

Schauder's Theorem: Key Concepts

Relative Compactness and Continuous Mappings



Theorem 5.55: Schauder's Theorem (Finite Dimension Version) I

Theorem 5.55 Let K be a nonempty, compact and convex subset of a Banach space X and let T be a continuous mapping of K into itself. Then T has a fixed point in K.

Proof Outline:

1. By Theorem 5.52: Any compact convex set K is homeomorphic to a compact convex subset C of some $\mathcal{H}_0$ (Hilbert cube)
2. By Theorem 5.54: C has the fixed point property
3. By Theorem 5.45: K has the fixed point property
4. Therefore T has a fixed point in K

Theorem 5.55: Schauder's Theorem (Finite Dimension Version) II

Theorem 5.56: Let K be a nonempty, compact and convex subset of a normed linear space X and let T be a continuous mapping of K into itself. Then T has a fixed point in K.

Theorem 5.57: Let T be a compact and continuous mapping of a normed linear space X into itself and let $T(X)$ be bounded. Then T has a fixed point.

Proof: Let K be the closed and convex hull of $T(X)$. Then K is bounded and $T(K) \subset K$. By Theorem 5.55 T has a fixed point in K. $\square$

Theorem 5.58: Reflexive Banach Spaces I

Theorem 5.58 Let X be a reflexive Banach space, K a closed and convex subset of X and T a weakly continuous mapping of K into a bounded subset of K. Then T has a fixed point in K.

Key Points:

- Extends Schauder to **reflexive** spaces
- Uses weak compactness instead of norm compactness
- Weakly continuous: If $x_n \rightharpoonup x$, then $Tx_n \rightharpoonup Tx$
- Applies to L^p spaces $(1 < p < \infty)$

Theorem 5.58: Reflexive Banach Spaces II

Theorem 5.63 (Potter [491]) Let X be a normed linear space. Let K be any closed and convex subset of X and T be a continuous mapping of K into a compact subset of X such that $T(\partial K) \subset K$. Then T has a fixed point.

Proof Idea:

- Define radial retraction R of X onto K
- Map $RT : K \to K$ is continuous into compact set
- By Schauder: RT has fixed point $u \in K$
- Can show this u is fixed point of T

Theorem 5.41: Lipschitzian Pseudocontractive Mappings

Theorem 5.41 (Kirk and Ray [334]) Let K be a closed and convex subset of a uniformly convex Banach space X. Let $F : K \to K$ be a Lipschitzian pseudocontractive mapping. Suppose for some $a \in K$, the set

$$G(a, Fa) = \{z \in K : \|z - a\| \geq \|z - Fa\|\}$$

is bounded. Then F has a fixed point in K.

Proof Strategy:

1. For each $y \in K$, define $F_y(x) = (1 - \alpha)y + \alpha F(F_y(y)), x \in K$
2. Show F_y is a contraction for appropriate $\alpha \in (0, 1)$
3. Obtain fixed point $T_a(y)$ of F_y
4. Prove that fixed point of T_a gives fixed point of F

Theorem 5.65 (Potter [491]) Let X be a normed linear space. Let K be any closed and convex subset of X and T be a continuous mapping of K into a compact subset of X such that $T(\partial K) \subset K$. Then T has a fixed point.

Geometric Interpretation:

- The boundary condition $T(\partial K) \subset K$ ensures:
    - No fixed point can lie on ∂K (boundary)
    - Continuous extension into interior
    - Topological degree argument applies
- Radial retraction $R(x) = \frac{x}{\max(1, \rho(x))}$ defines mapping RT
- RT inherits fixed point property from Schauder

Theorem 5.54 (Schauder-Tychonoff)
Let X be a locally convex topological vector space, K a nonempty compact convex subset of X, and $T : K \to K$ a continuous mapping. Then T has a fixed point in K.

Remark 5.25: This theorem generalizes Brouwer's result from finite-dimensional spaces to:

- Banach spaces (with compactness condition)
- Fréchet spaces
- Locally convex topological vector spaces
- Even infinite-dimensional normed spaces

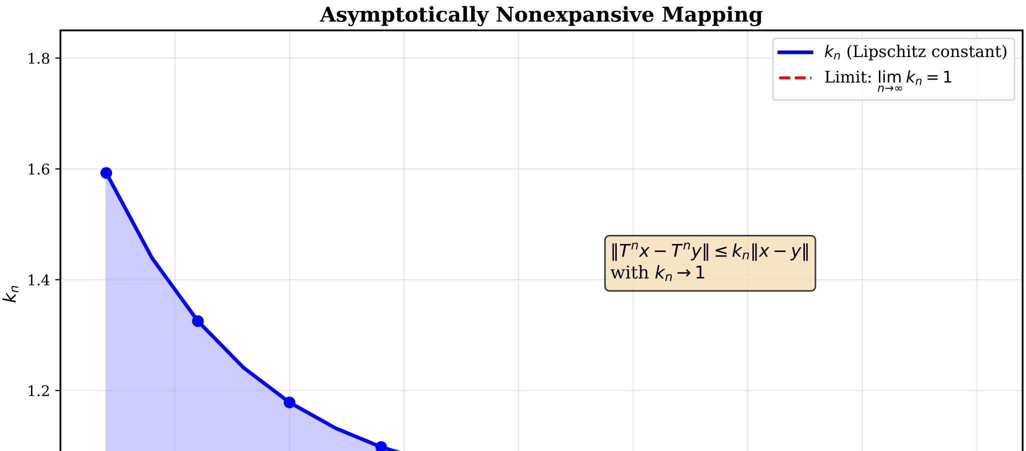
Lemma 5.8: Every continuous function $f : K \to K$ on finite-dimensional compact convex set has a fixed point.

- Extension step uses Tychonoff's compactness theorem
- Finite-dimensional projections inherit fixed points
- Limiting argument gives fixed point in full space

Asymptotically Nonexpansive Mappings: Detailed Analysis

Definition 5.24 and 5.27 A sequence $\{T_n\}$ of self-mappings of K is **asymptotically nonexpansive** if:

$$\|T_n x - T_n y\| \leq k_n \|x - y\| \quad \forall x, y \in K, \quad \lim_{n\to\infty} k_n = 1.$$



Theorem 5.44: Asymptotically Nonexpansive Sequences

Main Convergence Result

Theorem 5.44 Let X be a uniformly convex Banach space with Fréchet differentiable norm and K a closed and convex subset of X. Let C be a subset of K and $S = \{T_n\}_{n=1}^{\infty}$, an asymptotically nonexpansive sequence of self-mappings of K such that

1. $C \subset \bigcap_{n=1}^{\infty} \mathcal{F}(T_n)$
2. There exists $x_0 \in K$ for which $T_{n_k} x_0 \to z$ implies that $z \in C$
3. $T_n T_m x_0 - T_n x_0 \to 0$ as $n \to \infty$ for all (fixed) m

Then either:

1. $C = \emptyset$ and $\|T_n x_0\| \to +\infty$, or
2. $C \neq \emptyset$ and $T_n x_0$ converges weakly to an element of C.

Note: Hypothesis (c) is asymptotic regularity at x_0.

Numerical Example: Asymptotic Behavior

Python Implementation

```python
import numpy as np

# Asymptotically nonexpansive map
def T_n(x, n):
    """T_n(x) with k_n = 1 + exp(-n/10)"""
    k_n = 1.0 + np.exp(-n/10.0)
    # Example: contraction with varying rate
    return k_n * 0.5 * x

# Sequence of mappings
x0 = 1.0
x_history = [x0]
k_history = []

for n in range(1, 51):
    k_n = 1.0 + np.exp(-n/10.0)
    k_history.append(k_n)
    x_new = T_n(x_history[-1], n)
    x_history.append(x_new)

x_history = np.array(x_history)
k_history = np.array(k_history)

# Analysis
print("k_n values:", k_history[-5:])
```

Properties Observed:

- $k_n \to 1$ as $n \to \infty$
- Sequence $\{x_n\}$ converges
- Rate of convergence depends on k_n
- Slower than exponential for small n
- Eventually linear rate

Theorem 5.42 Guarantee: Under suitable conditions (uniformly convex Banach space, bounded set), asymptotically nonexpansive mappings have fixed points.

Summary: Fixed Point Theorems Hierarchy

Classification by Generality:

1. **Banach Contraction Mapping Principle**
   - Requires: $\|Tx - Ty\| \leq \lambda \|x - y\|$, $\lambda < 1$
   - Guarantees: Unique fixed point + computational scheme
2. **Brouwer's Fixed Point Theorem**
   - Requires: Continuous map, compact convex set in $\mathbb{R}^n$
   - Guarantees: Fixed point exists (no uniqueness, no algorithm)
3. **Schauder's Fixed Point Theorem**
   - Requires: Continuous map, compact convex set in Banach space
   - Guarantees: Fixed point exists in infinite dimensions
4. **Nonexpansive/Asymptotically Nonexpansive**
   - Requires: Lipschitz constant ≥ 1, special spaces
   - Guarantees: Fixed points under structural conditions

Applications and Significance

From Theory to Practice

Mathematical Applications:

- **Differential Equations:** Existence of solutions via fixed points of integral operators
- **Optimization:** Finding minimizers of convex functionals
- **Game Theory:** Nash equilibrium points (Kakutani's theorem uses fixed points)
- **Functional Analysis:** Extending results to infinite-dimensional spaces

Computational Methods:

- **Fixed point iteration:** $x_{n+1} = T(x_n)$
- **Krasnoselski iteration:** $x_{n+1} = (1 - \lambda)x_n + \lambda T(x_n)$
- **Mann iteration:** Weighted averages of iterates
- **Convergence analysis:** Error bounds from Lipschitz constants

Physical Applications:

- Bifurcation theory in dynamical systems
- Equilibrium solutions in mechanics
- Variational inequalities in PDEs

Open Problems and Extensions

Areas of Active Research:

1. **Computational Complexity:**
   - Efficient algorithms for finding fixed points
   - Approximation schemes for nonexpansive mappings
2. **Generalized Spaces:**
   - Fixed points in metric spaces with partial metrics
   - Fixed points in fuzzy metric spaces
   - Graph metric spaces
3. **Coupled Fixed Points:**
   - Systems of equations
   - Multiple interacting mappings
4. **Stability and Robustness:**
   - How do fixed points change under perturbations?
   - Stability of iteration schemes

Key References

From Pathak's Text

- Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
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- Kirk, W. A., & Schoenberg, R. (1972). Fixed point theorems for various classes of mappings.
- Goebel, K., & Kirk, W. A. (1972). A fixed point theorem for asymptotically nonexpansive mappings. *Proc. Amer. Math. Soc.*, 35, 171-174.
- Schauder, J. (1930). Der Fixpunktsatz in Funktionsräumen. *Studia Mathematica*, 2, 171-180.

Summary of Key Results

- **Brouwer's Theorem:** Continuous maps on compact convex sets in $\mathbb{R}^n$ have fixed points
- **Schauder's Theorem:** Extension to infinite-dimensional Banach spaces with appropriate compactness conditions
- **Nonexpansive Maps:** When $\|Tx - Ty\| \leq \|x - y\|$, fixed points exist under structural conditions
- **Asymptotically Nonexpansive:** $k_n \to 1$ provides intermediate theory between contractions and nonexpansive
- **Practical Impact:** These theorems guarantee existence (not always constructive) but motivate iterative algorithms

Questions?